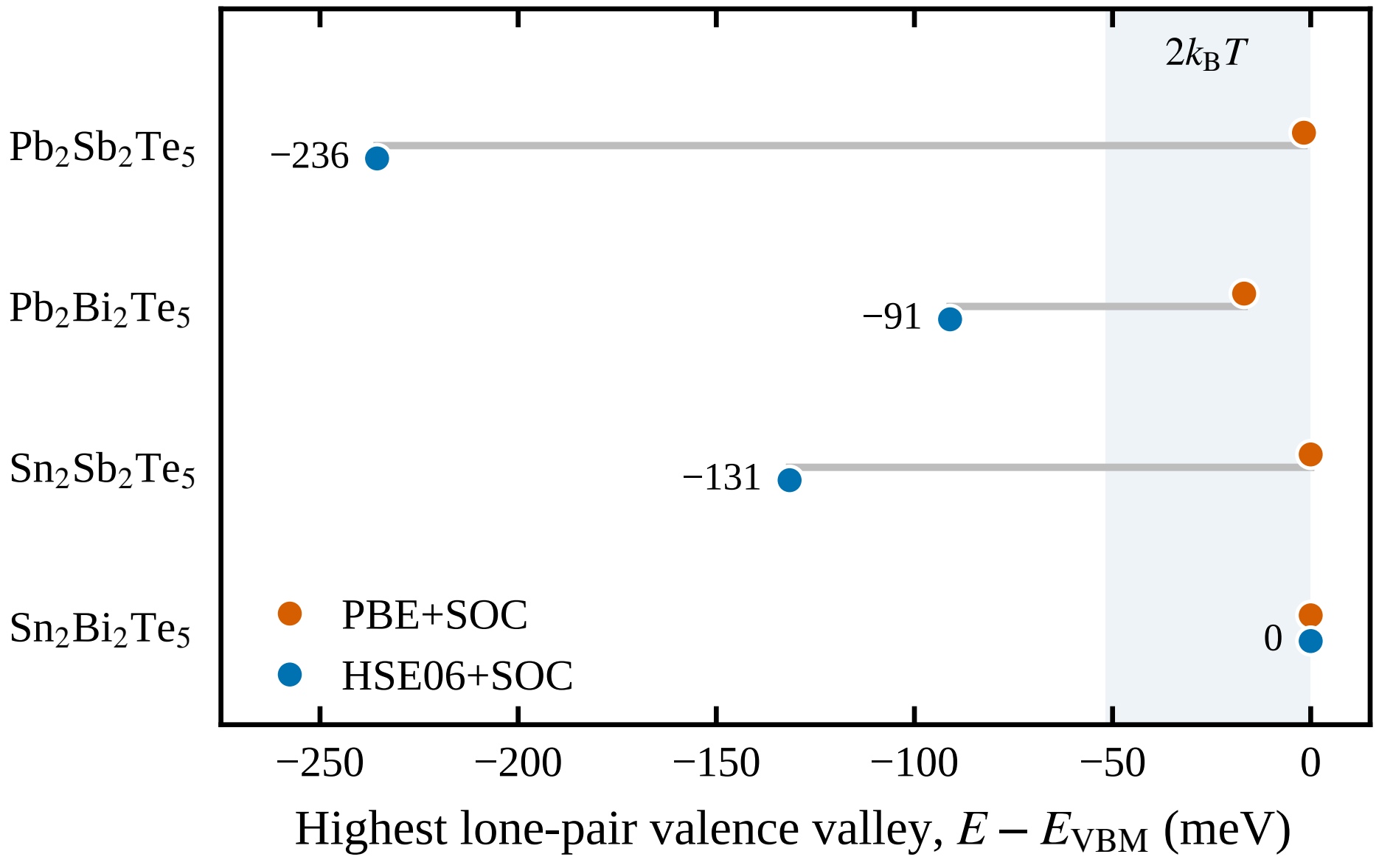
$2k_BT$
$Pb_2Sb_2Te_5$
$Pb_2Bi_2Te_5$
$Sn_2Sb_2Te_5$
$Sn_2Bi_2Te_5$
−236
−91
−131
0
PBE+SOC
HSE06+SOC
−250
−200
−150
−100
−50
0
Highest lone-pair valence valley, $E - E_{VBM}$ (meV)